K1 CSE3023 PEMBANGUNAN APLIKASI BERASASKAN WEB

SEM 4

LAB REPORT 2

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Lab Exercise

Source Code

Product.java

```java
package com.lab.model;

public class Product {

    private int id;
    private String name;
    private String category;
    private double price;
    private int quantity;

    //constructor kosong
    public Product(int id, String name, String category, double price, int quantity) {
        this.id = id;
        this.name = name;
        this.category = category;
        this.price = price;
        this.quantity = quantity;
    }

    //constructor tanpa id(insert)
    public Product(String name, String category, double price, int quantity) {
        this.name = name;
        this.category = category;
        this.price = price;
        this.quantity = quantity;
    }
```

```java
    //getter&setter
    public int getId() {
        return id;
    }

    public void setId(int id) {
        this.id = id;
    }

    public String getName() {
        return name;
    }

    public void setName(String name) {
        this.name = name;
    }

    public String getCategory() {
        return category;
    }

    public void setCategory(String category) {
        this.category = category;
    }

    public double getPrice() {
        return price;
    }

    public void setPrice(double price) {
        this.price = price;
    }
```

```java
    public int getQuantity() {
        return quantity;
    }

    public void setQuantity(int quantity) {
        this.quantity = quantity;
    }
}
```

```java
package com.lab.dao;

import com.lab.model.Product;
import java.sql.*;
import java.util.ArrayList;
import java.util.List;

public class productDAO {

    private String jdbcURL = "jdbc:mysql://localhost:3306/CSE3023";
    private String jdbcUsername = "root";
    private String jdbcPassword = "admin";

    private static final String INSERT_SQL
            = "INSERT INTO products (name, category, price, quantity) VALUES (?, ?, ?, ?)";

    private static final String SELECT_ALL
            = "SELECT * FROM products";

    private static final String SELECT_BY_ID
            = "SELECT * FROM products WHERE id=?";

    private static final String UPDATE_SQL
            = "UPDATE products SET name=?, category=?, price=?, quantity=? WHERE id=?";

    private static final String DELETE_SQL
            = "DELETE FROM products WHERE id=?";

    // connection
    private Connection getConnection() throws Exception {
        Class.forName("com.mysql.cj.jdbc.Driver");
        return DriverManager.getConnection(jdbcURL, jdbcUsername, jdbcPassword);
    }
```

```java
package com.lab.dao;

import com.lab.model.Product;
import java.sql.*;
import java.util.ArrayList;
import java.util.List;

public class productDAO {

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            = "UPDATE products SET name=?, category=?, price=?, quantity=? WHERE id=?";

    private static final String DELETE_SQL
            = "DELETE FROM products WHERE id=?";

    // connection
    private Connection getConnection() throws Exception {
        Class.forName("com.mysql.cj.jdbc.Driver");
        return DriverManager.getConnection(jdbcURL, jdbcUsername, jdbcPassword);
    }
```

InsertProductServlet.java

```java
// INSERT
public void insertProduct(Product p) {
    try (Connection conn = getConnection(); PreparedStatement ps = conn.prepareStatement(INSERT_SQL)) {

        ps.setString(1, p.getName());
        ps.setString(2, p.getCategory());
        ps.setDouble(3, p.getPrice());
        ps.setInt(4, p.getQuantity());
        ps.executeUpdate();

    } catch (Exception e) {
        e.printStackTrace();
    }
}

// SELECT ALL
public List<Product> getAllProducts() {
    List<Product> list = new ArrayList<>();

    try (Connection conn = getConnection(); PreparedStatement ps = conn.prepareStatement(SELECT_ALL); ResultSet rs = ps.executeQuery()) {

        while (rs.next()) {
            list.add(new Product(
                    rs.getInt("id"),
                    rs.getString("name"),
                    rs.getString("category"),
                    rs.getDouble("price"),
                    rs.getInt("quantity")
            ));
        }

    } catch (Exception e) {
        e.printStackTrace();
    }
    return list;
}
```

```java
// SELECT BY ID
public Product getProduct(int id) {
    Product p = null;

    try (Connection conn = getConnection(); PreparedStatement ps = conn.prepareStatement(SELECT_BY_ID)) {

        ps.setInt(1, id);
        ResultSet rs = ps.executeQuery();

        if (rs.next()) {
            p = new Product(
                    rs.getInt("id"),
                    rs.getString("name"),
                    rs.getString("category"),
                    rs.getDouble("price"),
                    rs.getInt("quantity")
            );
        }

    } catch (Exception e) {
        e.printStackTrace();
    }
    return p;
}
```

ViewProductServlet.java

```java
@Override
protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

    response.setContentType("text/html");
    PrintWriter out = response.getWriter();

    List<Product> list = dao.getAllProducts();

    out.println("<h2>Product List</h2>");
    out.println("<a href='add_product.html'>Add Product</a><br><br>");

    out.println("<table border='1'>");
    out.println("<tr><th>ID</th><th>Name</th><th>Category</th><th>Price</th><th>Qty</th><th>Action</th></tr>");

    for (Product p : list) {
        out.println("<tr>");
        out.println("<td>" + p.getId() + "</td>");
        out.println("<td>" + p.getName() + "</td>");
        out.println("<td>" + p.getCategory() + "</td>");
        out.println("<td>" + p.getPrice() + "</td>");
        out.println("<td>" + p.getQuantity() + "</td>");

        out.println("<td>"
            + "<a href='UpdateProductServlet?id=" + p.getId() + "'>Edit</a> | "
            + "<a href='DeleteProductServlet?id=" + p.getId() + "' "
            + "onclick=\"return confirm('Are you sure?')\">Delete</a>"
            + "</td>");

        out.println("</tr>");
    }

    out.println("</table>");
}
```

UpdateProductServlet.java

```java
@Override
protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

    int id = Integer.parseInt(request.getParameter("id"));
    Product p = dao.getProduct(id);

    response.setContentType("text/html");
    PrintWriter out = response.getWriter();

    out.println("<h2>Update Product</h2>");
    out.println("<form method='post'>");

    out.println("<input type='hidden' name='id' value='" + p.getId() + "'>");
    out.println("Name: <input type='text' name='name' value='" + p.getName() + "'><br><br>");
    out.println("Category: <input type='text' name='category' value='" + p.getCategory() + "'><br><br>");
    out.println("Price: <input type='text' name='price' value='" + p.getPrice() + "'><br><br>");
    out.println("Quantity: <input type='text' name='quantity' value='" + p.getQuantity() + "'><br><br>");

    out.println("<input type='submit' value='Update'>");
    out.println("</form>");
}
```

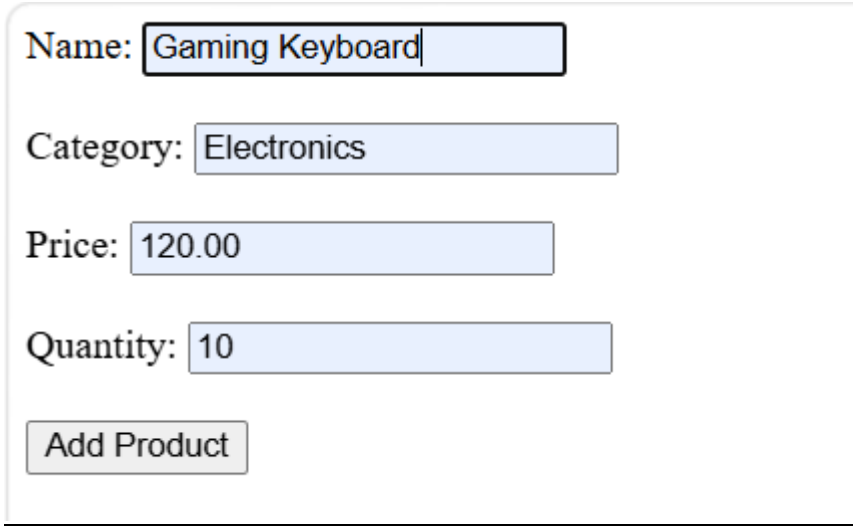
```java
@Override
protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

    int id = Integer.parseInt(request.getParameter("id"));
    String name = request.getParameter("name");
    String category = request.getParameter("category");
    double price = Double.parseDouble(request.getParameter("price"));
    int quantity = Integer.parseInt(request.getParameter("quantity"));

    Product p = new Product(id, name, category, price, quantity);

    dao.updateProduct(p);

    response.sendRedirect("ViewProductServlet");
}
```

DeleteProductServlet.java

```java
@Override
protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

    // ambil id dari URL
    int id = Integer.parseInt(request.getParameter("id"));

    // call DAO delete
    dao.deleteProduct(id);

    // redirect balik ke list
    response.sendRedirect("ViewProductServlet");
}

// optional (kalau POST pun nak support)
@Override
protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

    doGet(request, response);
}
```

Output

localhost:8080/LabExercise/add_product.html
Name:
Category:
Price:
Quantity:
Add Product


Name: Gaming Keyboard
Category: Electronics
Price: 120.00
Quantity: 10
Add Product

Product List

Add Product

ID	Name	Category	Price	Qty	Action
2050	mouse	electronics	1200.0	3	Edit | Delete
2051	Gaming Keyboard	Electronics	120.0	10	Edit | Delete

Update Product

Name: mouse

Category: electronics

Price: 1200.0

Quantity: 3

Update

Product List

Add Product

ID	Name	Category	Price	Qty	Action
2050	mouse	electronics	1200.0	3	Edit | Delete
2051	Gaming Keyboard	Electronics	120.0	10	Edit | Delete

localhost:8080 says

Are you sure?

OK Cancel